

Research Notes on Learning rate neural network

Learning rate neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

Deep Neural Network Estimations Deep neural networks can be used to estimate the entropy of a stochastic process through an arrangement called a Neural Joint Entropy Estimator (NJEE). Such an estimation provides insights on the effects of input random variables on an independent random variable. Practically, the DNN is trained as a classifier that maps an input vector or matrix X to an output probability distribution over the possible classes of random variable Y , given input X . For example, in image classification tasks, the NJEE maps a vector of pixels' color values to probabilities over possible image classes. In practice, the probability distribution of Y is obtained by a Softmax layer with number of nodes that is equal to the alphabet size of Y . NJEE uses continuously differentiable activation functions, such that the conditions for the universal approximation theorem holds. It is shown that this method provides a strongly consistent estimator and outperforms other methods in cases of large alphabet sizes.

>> Section 2

2000s Neural networks entered a lull, and simpler models that use task-specific handcrafted features such as Gabor filters and support vector machines (SVMs) became the preferred choices in the 1990s and 2000s, because of artificial neural networks' computational cost and a lack of understanding of how the brain wires its biological networks. In 2003, LSTM became competitive with traditional speech recognizers on certain tasks. In 2006, Alex Graves, Santiago Fernández, Faustino Gomez, and Schmidhuber combined it with connectionist temporal classification (CTC) in stacks of LSTMs. In 2009, it became the first RNN to win a pattern recognition contest, in connected handwriting recognition. In 2006, publications by Geoff Hinton, Ruslan Salakhutdinov, Osindero and Teh deep belief networks were developed for generative modeling. They are trained by training one restricted Boltzmann machine, then freezing it and training another one on top of the first one, and so on, then optionally fine-tuned using supervised backpropagation. They could model high-dimensional probability distributions, such as the distribution of MNIST images, but convergence was slow. The impact of deep learning in industry began in the early 2000s, when CNNs already processed an estimated 10% to 20% of all the checks written in the US, according to Yann LeCun. Industrial applications of deep learning to

large-scale speech recognition started around 2010. The 2009 NIPS Workshop on Deep Learning for Speech Recognition was motivated by the limitations of deep generative models of speech, and the possibility that given more capable hardware and large-scale data sets that deep neural nets might become practical. It was believed that pre-training DNNs using generative models of deep belief nets (DBN) would overcome the main difficulties of neural nets. However, it was discovered that replacing pre-training with large amounts of training data for straightforward backpropagation when using DNNs with large, context-dependent output layers produced error rates dramatically lower than then-state-of-the-art Gaussian mixture model (GMM)/Hidden Markov Model (HMM) and also than more-advanced generative model-based systems. The nature of the recognition errors produced by the two types of systems was characteristically different, offering technical insights into how to integrate deep learning into the existing highly efficient, run-time speech decoding system deployed by all major speech recognition systems. Analysis around 2009–2010, contrasting the GMM (and other generative speech models) vs. DNN models, stimulated early industrial investment in deep learning for speech recognition. That analysis was done with comparable performance (less than 1.5% in error rate) between discriminative DNNs and generative models. In 2010, researchers extended deep learning from TIMIT to large vocabulary speech recognition, by adopting large output layers of the DNN based on context-dependent HMM states constructed by decision trees.

>> Section 3

Scale-up/out and accelerated DNN training and decoding Sequence discriminative training Feature processing by deep models with solid understanding of the underlying mechanisms Adaptation of DNNs and related deep models Multi-task and transfer learning by DNNs and related deep models CNNs and how to design them to best exploit domain knowledge of speech RNN and its rich LSTM variants Other types of deep models including tensor-based models and integrated deep generative/discriminative models. More recent speech recognition models use Transformers or Temporal Convolution Networks with significant success and widespread applications. All major commercial speech recognition systems (e.g., Microsoft Cortana, Xbox, Skype Translator, Amazon Alexa, Google Now, Apple Siri, Baidu and iFlyTek voice search, and a range of Nuance speech products, etc.) are based on deep learning.

>> Section 4

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Artificial neural networks (ANNs) or connectionist systems are computing systems inspired by the biological neural networks that constitute animal brains. Such systems learn (progressively improve their ability) to do tasks by considering examples, generally without task-specific programming. For example, in image recognition, they might learn to identify images that contain cats by analyzing example images that have been manually labeled as "cat" or "no cat" and using the analytic results to identify cats in other images. They have found most use in applications difficult to express with a traditional computer algorithm using rule-based programming. An ANN is based on a collection of connected units called artificial neurons, (analogous to biological neurons in a biological brain). Each connection (synapse) between neurons can transmit a signal to another neuron. The receiving (postsynaptic) neuron can process the signal(s) and then signal downstream neurons connected to it. Neurons may have state, generally represented by real numbers, typically between 0 and 1. Neurons and synapses may also have a weight that varies as learning proceeds, which can increase or decrease the strength of the signal that it sends downstream. Typically, neurons are organized in layers. Different layers may perform different kinds of transformations on their inputs. Signals travel from the first (input), to the last (output) layer, possibly after traversing the layers multiple times. The original goal of the neural network approach was to solve problems in the same way that a human brain would. Over time, attention focused on matching specific mental abilities, leading to deviations from biology such as backpropagation, or passing information in the reverse direction and adjusting the network to reflect that information. Neural networks have been used on a variety of tasks, including computer vision, speech recognition, machine translation, social network filtering, playing board and video games and medical diagnosis. As of 2017, neural networks typically have a few thousand to a few million units and millions of connections. Despite this number being several order of magnitude less than the number of neurons on a human brain, these networks can perform many tasks at a level beyond that of humans (e.g., recognizing faces, or playing "Go").